Financial Engineering - Assignment RM 5

Group 6

Affine form of the discount factor

We start from the forward discount factor formula given in the OU framework:

$$\frac{B(t,T_1,T_2)}{B(t_0,T_1,T_2)} = \exp\left(-x_t\left[\sigma(t,T_2)-\sigma(t,T_1)\right] - \frac{1}{2}\int_{t_0}^{t}\left[\sigma^2(u,T_2)-\sigma^2(u,T_1)\right]du\right)$$

where $\sigma(u,\tau) = \dfrac{\sigma}{a}\left(1-e^{-a(\tau-u)}\right)$.

Set $T_1 = s$, $T_2 = \tau$, and evaluate at $t = s$:

$$\frac{B(s,s,\tau)}{B(t_0,s,\tau)} = \exp\left(-x_s\left[\sigma(s,\tau)-\sigma(s,s)\right] - \frac{1}{2}\int_{t_0}^{s}\left[\sigma^2(u,\tau)-\sigma^2(u,s)\right]du\right)$$

Since $\sigma(s,s) = \dfrac{\sigma}{a}\left(1-e^{-a(s-s)}\right) = 0$, and noting that $B(s,s,\tau) = B(s,\tau)$ (spot discount factor at time s):

$$\frac{B(s,\tau)}{B(t_0,s,\tau)} = \exp\left(-x_s\,\sigma(s,\tau) - \frac{1}{2}\int_{t_0}^{s}\left[\sigma^2(u,\tau)-\sigma^2(u,s)\right]du\right)$$

Using the definition of the forward discount factor $B(t_0,s,\tau) = \dfrac{B(t_0,\tau)}{B(t_0,s)}$:

$$B(s,\tau) = \frac{B(t_0,\tau)}{B(t_0,s)}\exp\left(-\frac{1}{2}\int_{t_0}^{s}\left[\sigma^2(u,\tau)-\sigma^2(u,s)\right]du\right)\cdot\exp\left(-x_s\,\sigma(s,\tau)\right)$$

Identifying $C(s,\tau) = \sigma(s,\tau) = \dfrac{1-e^{-a(\tau-s)}}{a}$ and:

$$A(s,\tau) = \frac{B(t_0,\tau)}{B(t_0,s)}\exp\left(-\frac{1}{2}\int_{t_0}^{s}\left[\sigma^2(u,\tau)-\sigma^2(u,s)\right]du\right)$$

we obtain the affine form:

$$\boxed{B(s,\tau) = A(s,\tau)\,e^{-x_s C(s,\tau)}}$$

Results and Discussion

Base Case: EPE and PFE

The base case corresponds to a vanilla 10-year payer IRS entered on February 15th, 2008, with no credit mitigation. The EPE and PFE profiles exhibit a characteristic sawtooth pattern throughout the life of the swap: exposure peaks just before each annual coupon payment date, when the present value of remaining fixed cashflows is highest, and drops sharply immediately after each payment, as one coupon is removed from the remaining cashflow schedule. This pattern is a direct consequence of the discrete coupon structure of the IRS and is economically expected. The sawtooth is more pronounced in the EPE than in the PFE, since the latter is driven by the tail of the distribution and is less sensitive to the average behaviour around coupon dates.

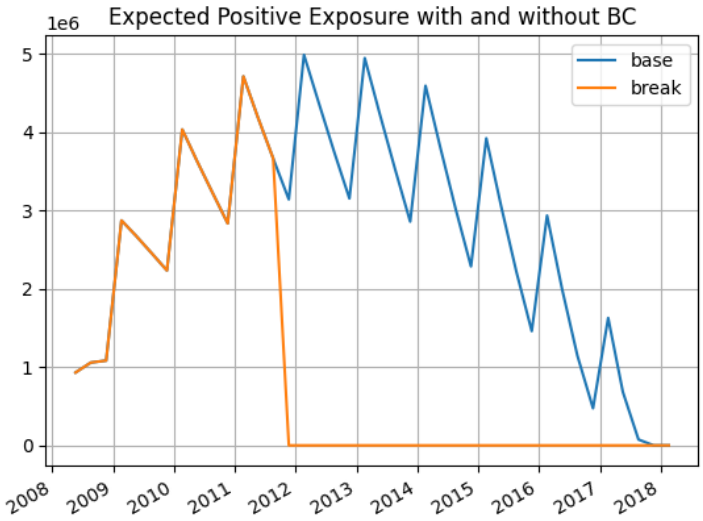
The smooth envelope is the result of two competing mechanisms acting on different time scales:

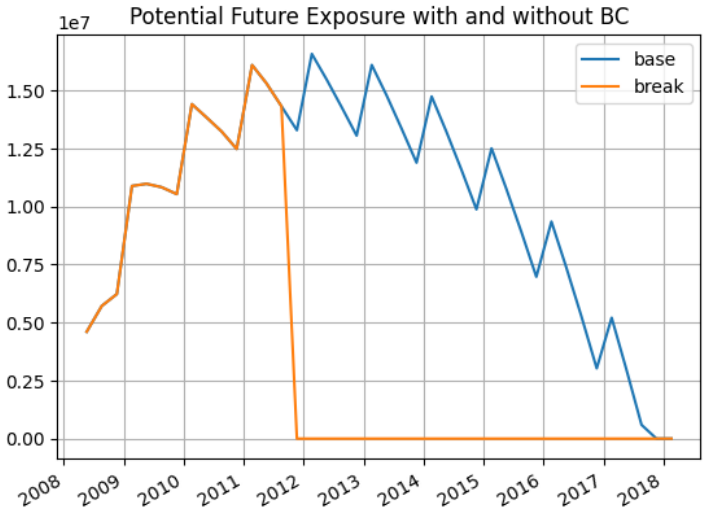
- **Diffusion effect.** The variance of the short rate process x_t grows as $\frac{\sigma^2}{2a}\left(1 - e^{-2at}\right)$, i.e. roughly with $\sqrt{t}$ in the early life of the swap. This widens the distribution of the future MtM and pushes both EPE and PFE upwards.
- **Amortisation effect.** As time elapses, the residual tenor of the swap shrinks. Fewer remaining cashflows mean that even large rate moves translate into smaller MtM impact, which pulls both EPE and PFE downwards.

The interplay of the two effects generates the typical hump shape, peaking around mid-life of the swap. The Peak-PFE at 95% reaches approximately 1.65×10^7 EUR.

Break Clause at 3.5 Years

The introduction of a mandatory break clause at 3.5 years (mid-2011) eliminates all exposure beyond that date. The EPE and PFE break curves are identical to the base case up to the break date, then collapse to zero immediately after, since the position is assumed terminated upon exercise. This confirms the effectiveness of break clauses as a duration-limiting risk mitigation tool. The Peak-PFE under the break clause scenario is significantly lower than the base case, bounded by the maximum exposure achievable within the 3.5-year window.





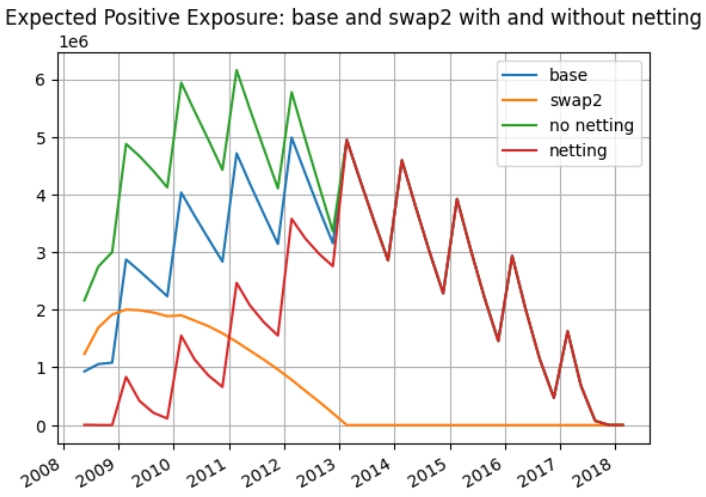
Netting Agreement

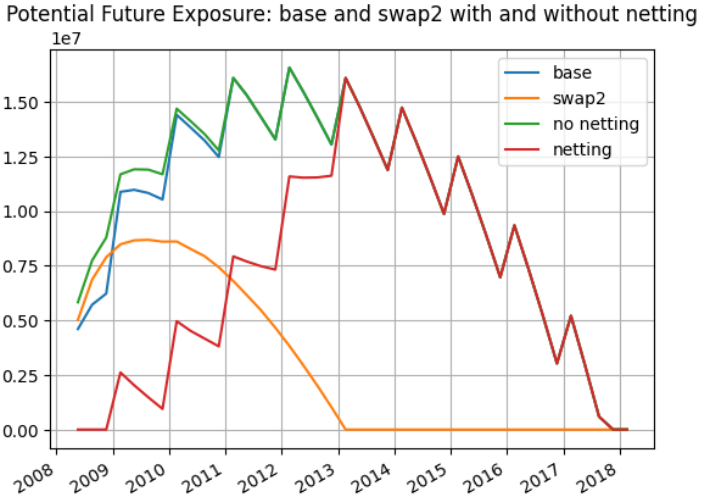
The netting analysis compares four quantities at each simulation date: the standalone exposure of swap 1 (base), the standalone exposure of swap 2, their sum (no netting), and the positive part of their combined MtF (netting). The theoretical inequality

$$\mathrm{EPE}_{\text{netting}}(s) \leq \mathrm{EPE}_{\text{no netting}}(s)$$

is verified, since $\mathbb{E}[\max(V_1 + V_2, 0)] \leq \mathbb{E}[\max(V_1, 0)] + \mathbb{E}[\max(V_2, 0)]$.

The netted EPE lies *above* the standalone base curve over the $[0, 5\text{y}]$ window. This is not due to a simultaneous in-the-money condition (the payer and the receiver are directionally opposite in rates by construction), but to the *notional asymmetry*: the receiver carries 150 M EUR against the 100 M EUR of the payer, so in rate-falling scenarios the netted MtM is dominated by the receiver and exceeds in magnitude what the payer alone produces in symmetric rate-rising scenarios. Sub-additivity with respect to the no-netting sum is nevertheless preserved. After swap 2 matures (around 2013), the netted and no-netting curves collapse back onto the base curve.





CSA Agreement with Annual Margin Calls

The collateral parameters are chosen asymmetrically in favor of Polimi Bank (party B):

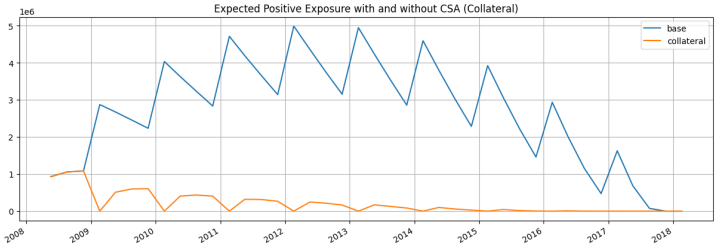
- $\mathrm{THR}_B \to +\infty$: by design, B is never required to post collateral;
- $\mathrm{THR}_C = 0$, $\mathrm{MTA}_C = 0$: the counterparty posts as soon as any positive exposure arises;
- $\mathrm{IA}_B = \mathrm{IA}_C = 0$: no independent amount.

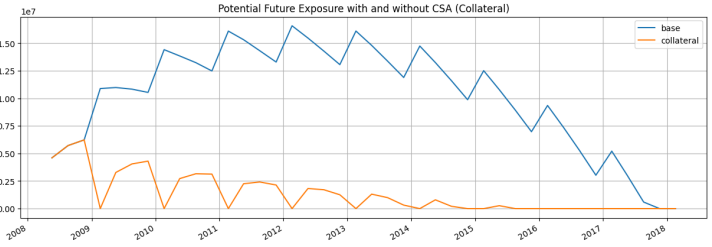
The bilateral CSA logic is implemented in `update_collateral`, which computes a scenario-wise target variation margin

$$\mathrm{VM}^* \;=\; \max(\mathrm{MtM} - \mathrm{THR}_C, 0) \cdot \mathbf{1}_{\mathrm{MtM}\geq 0} \;+\; \min(\mathrm{MtM} + \mathrm{THR}_B, 0) \cdot \mathbf{1}_{\mathrm{MtM}<0},$$

applies the MTA filter on $\Delta\mathrm{VM} = \mathrm{VM}^* - \mathrm{VM}$, and finally enforces the caps. With the parameters above this collapses to "the counterparty posts collateral equal to $\max(\mathrm{MtM}, 0)$ at every annual posting date", while Polimi Bank never posts. Between two posting dates the posted collateral is remunerated at the shortest-tenor zero rate available on the simulated HW curve, used as a proxy for the path-wise risk-free rate.

The resulting EPE with collateral is much lower at every simulation date, confirming the effectiveness of the CSA. The residual exposure visible in the plots arises from the *margin period of risk*: between two consecutive posting dates the MtM moves freely while the posted collateral is frozen (modulo risk-free remuneration), so the exposure can re-accumulate. The size of the residual exposure scales approximately as $\sigma\sqrt{\Delta t_{\mathrm{call}}}$ with the inter-call window Δt_{call}: with monthly, weekly or daily margining, the residual would shrink by a factor $\sqrt{12}$, $\sqrt{52}$, $\sqrt{252}$ respectively. The PFE with collateral remains well below 10^6 throughout, compared with a Peak-PFE of 1.65×10^7 in the base case, a reduction of more than 90%.





CVA Estimation

Under the assumptions of unilateral default risk, independence between exposure and default, flat hazard rate $\lambda = 200$ bps and LGD = 60%, the CVA is computed as:

$$\text{CVA} = \text{LGD} \sum_{i=1}^{N} \text{EPE}(t_i) \cdot B(0, t_i) \cdot \left(e^{-\lambda t_{i-1}} - e^{-\lambda t_i} \right)$$

Scenario	CVA (EUR)	CVA (M EUR)
Base	251,722	0.252
Break clause	106,107	0.106
Swap 2 standalone	71,929	0.072
No netting	323,651	0.324
Netting	164,570	0.165
Collateral (CSA)	22,152	0.022

Table 1: CVA estimates across mitigation scenarios.

The no-netting CVA (323,651) equals the sum of the two standalone CVAs (251,722 + 71,929), confirming CVA additivity without netting. The netting agreement reduces CVA by 49%, the break clause by 58%, and the CSA by 91%. The residual CVA under the CSA reflects the margin period of risk between annual margin calls.

CVA Sensitivity Analysis

Three first-order sensitivities are computed by bump-and-reprice (+1bp), targeting the main market risk factors of the CVA. The selected sensitivities capture the main drivers of CVA: counterparty credit risk (CR Delta), interest-rate risk (IR Delta), and model volatility risk (Vega).

Sensitivity	Risk factor	CVA bumped (EUR)	ΔCVA / bp (EUR)
CR Delta	Hazard rate λ	252,875	+1,152
IR Delta	Parallel curve shift	219,472	−32,251
Vega	HW volatility σ	210,331	−41,392

Table 2: CVA sensitivities (base CVA = EUR 251,722, bump = +1bp).

CR Delta. A positive sensitivity is expected: a higher hazard rate λ increases the marginal default probability $e^{-\lambda t_{i-1}} - e^{-\lambda t_i}$ at every simulation date, mechanically inflating the CVA integral. The numerical magnitude (+1,152 EUR per bp) is moderate relative to the base CVA of 251,722 EUR, reflecting the already significant default intensity at the baseline level $\lambda = 200$ bps.

IR Delta. Our implementation captures two channels simultaneously: after the +1 bp parallel shift on the zero-rate curve we recompute the affine reconstruction and re-evaluate the EPE via `compute_epe_base`, holding the par rate K frozen at its inception value. The *discounting* channel reduces the present value of future losses; the *exposure* channel reflects the residual mismatch from the swap no longer being at par on the shifted curve. The two effects pull in opposite directions; numerically the second dominates, yielding a CVA decrease of about 32,000 EUR per bp.

Vega. The Hull–White volatility σ controls the dispersion of x_t and, through $B(s,\tau) = A(s,\tau)\exp(-x_s C(s,\tau))$, that of the simulated discount-factor curve. Re-running the Monte Carlo with a fresh draw of normals would inject sampling noise of order $\mathcal{O}(1/\sqrt{N})$, comparable to the signal for a 1 bp bump on $\sigma = 94$ bp.

We therefore use common random numbers, reseeding `numpy` before regenerating `x_paths` so that the same realisations drive base and bumped simulations: the reported Vega (−41,392 EUR per bp) is the estimator bias rather than its Monte Carlo noise floor. The negative sign reflects the fact that a wider x_s distribution lowers the average PV of the residual fixed leg more than it raises the average payer MtM.